

MINT CLASSICS – INVENTORY OPTIMIZATION

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About the company: Mint Classics Company is a retailer of classic model cars and other vehicles

Problem Description: The Company is looking for suggestions and recommendations for reorganizing or reducing inventory, while still maintaining timely service to their customers.

We seek to provide answers for the following questions:

1. Is there a way to reduce inventory without affecting sales
2. Is the company keeping excess/lower stock of any inventory
3. Where are the items being stored
4. Can the company reduce the number of warehouses

WAREHOUSE REARRANGEMENT

We assume the variable WarehousePctCap as Ware House Percentage Capacity indicating how much available space is currently being used by the respective warehouses.

warehousename	warehousePctCap	quantity
South	75	79380
West	50	124880
North	72	131688
East	67	219183

Table 1: Inventory currently in each warehouse

No warehouse is working at maximum capacity. Although the South warehouse is currently at a greater capacity, it stores the least quantity of stock, indicating it to be a smaller warehouse.

We look at the kind of vehicles being stored in each warehouse

productline	warehouse	quantityInStock
Motorcycles	North	69401
Planes	North	62287
Classic Cars	East	219183
Vintage Cars	West	124880
Trucks and Buses	South	35851
Ships	South	26833
Trains	South	16696

Table 2: Product lines across warehouses

With respect to the product lines, it can be observed that the warehouses are mutually exclusive i.e. the same product line is not spread across different warehouses. This makes it easier for the company to carry out the operations. While changing the warehouse arrangement, care should be taken that this mutual exclusivity is maintained.

We calculate the maximum capacity of each warehouse based on its current percentage occupancy and items already in stock. To avoid congestions and other inefficiencies, we propose a maximum 85% utilization capacity.

warehousename	TotalQuantityInStock	warehousePctCap	max_capacity	85PctCapacity	max_remaining_capacity	85Pct_remaining_capacity
East	219183	67	327139	278068	107956	58885
North	131688	72	182900	155465	51212	23777
West	124880	50	249760	212296	124880	87416
South	79380	75	105840	89964	26460	10584

Table 3: Calculating the optimum and available capacity of each warehouse

It can be seen that the inventory in South warehouse in its entirety (79380) can be reallocated towards the other 3 warehouses while still being under the 85% threshold for each warehouse. SUGGESTION - to accommodate the Trains to the North, Ships to the east and the trucks and buses to the west.

INVENTORY ANALYSIS

We check whether the company has sufficient inventory for all the products ordered as determine whether there exists a scope to reduce the inventory without affecting the profits of the company.

productName	quantityInStock	quantityOrdered	ExcessStock
1985 Toyota Supra	7733	NULL	NULL
1960 BSA Gold Star DBD34	15	1015	-1000
1968 Ford Mustang	68	933	-865
1997 BMW F650 ST	178	1014	-836
1928 Ford Phaeton Deluxe	136	972	-836
Pont Yacht	414	958	-544
F/A 18 Hornet 1/72	551	1047	-496
2002 Yamaha YZR M1	600	992	-392
1928 Mercedes-Benz SSK	548	880	-332
1911 Ford Town Car	540	832	-292
1996 Peterbilt 379 Stake ...	814	988	-174
The Mayflower	737	898	-161
Diamond T620 Semi-Skirte...	1016	979	37
P-51-D Mustang	992	917	75
1969 Ford Falcon	1049	965	84

Table 4: Bottom 15 products with respect to excess capacity (excluding the first entry)

The above table gives details of those products, for whom the warehouse fails to meet its demand. However, it is to be noted that 1985 Toyota Supra indicates Null value, since no order has come up for it. Hence it is suggested that the company drastically reduce the inventory by selling these model cars at a discount or reduced profit.

There are 11 models for which demand is more than the existing quantity in stock (Indicated by the negative Excess Stock).

productName	quantityInStock	quantityOrdered	ExcessStock
1968 Ford Mustang	68	933	-865
1911 Ford Town Car	540	832	-292
1928 Mercedes-Benz SSK	548	880	-332
1960 BSA Gold Star DBD34	15	1015	-1000
1997 BMW F650 ST	178	1014	-836
1996 Peterbilt 379 Stake Bed with Outrigger	814	988	-174
1928 Ford Phaeton Deluxe	136	972	-836
2002 Yamaha YZR M1	600	992	-392
The Mayflower	737	898	-161
F/A 18 Hornet 1/72	551	1047	-496
Pont Yacht	414	958	-544

Table 5: Products with demand more than stock available

Suggestion : Also, to maintain sufficient stock to ensure timely dispatch, it is advisable to maintain a certain percentage of excess stock, this should be based on the demand for each product, Higher the orders placed more the excess stock can be maintained. We propose the following criteria: Top 25% ordered models can maintain 200% excess quantity, the next 25% can maintain 150% excess quantity, the next 25% with 100 % excess quantity while the bottom 25% can maintain 50% excess quantity. By % excess quantity, we mean the excess quantity as a percentage of total quantity ordered.

productName	quantityInStock	quantityOrdered	ExcessStock	ExcessPct	Rnk	ExcessPctRequired	ExcessSugg	Action_
1992 Ferrari 360 Spider red	8347	1808	6539	361.6704	1	200	3616	Reduce Stock
1937 Lincoln Berline	8693	1111	7582	682.4482	2	200	2222	Reduce Stock
American Airlines: MD-11S	8820	1085	7735	712.9032	3	200	2170	Reduce Stock
1941 Chevrolet Special Deluxe Cabriolet	2378	1076	1302	121.0037	4	200	2152	Increase Stock
1930 Buick Marquette Phaeton	7062	1074	5988	557.5419	5	200	2148	Reduce Stock
1940s Ford truck	3128	1061	2067	194.8162	6	200	2122	Increase Stock
1969 Harley Davidson Ultimate Chopper	7933	1057	6876	650.5203	7	200	2114	Reduce Stock
1957 Chevy Pickup	6125	1056	5069	480.0189	8	200	2112	Reduce Stock
1964 Mercedes Tour Bus	8258	1053	7205	684.2355	9	200	2106	Reduce Stock
1956 Porsche 356A Coupe	6600	1052	5548	527.3764	10	200	2104	Reduce Stock
Corsair F4U (Bird Cage)	6812	1051	5761	548.1446	11	200	2102	Reduce Stock
F/A 18 Hornet 1/72	551	1047	-496	-47.3734	12	200	2094	Increase Stock
1980s Black Hawk Helicopter	5330	1040	4290	412.5000	13	200	2080	Reduce Stock
1913 Ford Model T Speedster	4189	1038	3151	303.5645	14	200	2076	Reduce Stock
1997 BMW R 1100 S	7003	1033	5970	577.9284	15	200	2066	Reduce Stock
1972 Alfa Romeo GTA	3252	1030	2222	215.7282	16	200	2060	Reduce Stock
1962 Volkswagen Microbus	2327	1029	1298	126.1419	17	200	2058	Increase Stock
2002 Suzuki XREO	9997	1028	8969	872.4708	18	200	2056	Reduce Stock
The USS Constitution Ship	7083	1020	6063	594.4118	19	200	2040	Reduce Stock

Table 6: Action to be taken

There exist 24 items for which the stock should be increased while for the remaining, the inventory should be decreased.

PROFIT ANALYSIS

The Top 15 and Bottom 15 profitable products are as follows

productName	productLine	Sales	cost	Profit
1992 Ferrari 360 Spider red	Classic Cars	271014.78	137727.20	133287.58
1952 Alpine Renault 1300	Classic Cars	190017.96	94735.38	95282.58
2001 Ferrari Enzo	Classic Cars	182439.52	93009.07	89430.45
2003 Harley-Davidson Eagle Drag Bike	Motorcycles	170686.00	89654.70	81031.30
1968 Ford Mustang	Classic Cars	157749.08	86664.06	71085.02
1969 Ford Falcon	Classic Cars	145082.38	76489.05	68593.33
1928 Mercedes-Benz SSK	Vintage Cars	132275.98	63852.80	68423.18
2002 Suzuki XREO	Motorcycles	135767.03	68125.56	67641.47
1917 Grand Touring Sedan	Vintage Cars	140535.60	79590.60	60945.00
1948 Porsche Type 356 Roadster	Classic Cars	117317.49	57000.72	60316.77
1957 Corvette Convertible	Classic Cars	130749.31	70839.09	59910.22
1980s Black Hawk Helicopter	Planes	131130.54	73251.96	57878.58
1962 Volkswagen Microbus	Trucks and ...	118774.33	63118.86	55655.47
1976 Ford Gran Torino	Classic Cars	121890.60	67243.35	54647.25
1932 Model A Ford J-Coupe	Vintage Cars	109992.01	55965.36	54026.65

Table 7: Top Profitable Models

productName	productLine	Sales	cost	Profit	profitpct
1939 Chevrolet Deluxe Coupe	Vintage Cars	28052.94	21148.09	6904.85	32.649993
Boeing X-32A JSF	Planes	40108.22	29460.23	10647.99	36.143608
1982 Ducati 996 R	Motorcycles	33268.76	21870.84	11397.92	52.114688
1930 Buick Marquette Phaeton	Vintage Cars	38377.22	26816.46	11560.76	43.110686
1936 Mercedes Benz 500k Roadster	Vintage Cars	29763.39	17922.00	11841.39	66.071811
1982 Ducati 900 Monster	Motorcycles	57995.25	44603.70	13391.55	30.023406
1970 Chevy Chevelle SS 454	Classic Cars	53236.67	39539.72	13696.95	34.640989
1948 Porsche 356-A Roadster	Classic Cars	64897.14	51151.10	13746.04	26.873401
1971 Alpine Renault 1600s	Classic Cars	51119.85	36458.10	14661.75	40.215343
Pont Yacht	Ships	44659.83	29870.10	14789.73	49.513493
1966 Shelby Cobra 427 S/C	Classic Cars	42015.54	26699.70	15315.84	57.363341
1982 Lamborghini Diablo	Classic Cars	29583.27	14161.28	15421.99	108.902514
1962 City of Detroit Streetcar	Trains	50942.77	35428.05	15514.72	43.792193
1958 Chevy Corvette Limited Edition	Classic Cars	31627.96	15639.53	15988.43	102.230885
1911 Ford Town Car	Vintage Cars	45306.77	27705.60	17601.17	63.529286

Table 8: Least Profitable Models

These are based on orders not cancelled by the customer. It can be viewed that Classic Cars Take up half of the top spots.

productLine	Sales	cost	Profit	profitpct	ProfitShare
Classic Cars	3774647.39	2279989.85	1494657.54	65.555447	40.06
Vintage Cars	1755309.57	1035648.14	719661.43	69.488990	19.29
Motorcycles	1117515.62	649159.42	468356.20	72.148102	12.55
Trucks and Buses	1024113.57	623560.35	400553.22	64.236480	10.73
Planes	903119.39	557223.35	345896.04	62.074936	9.27
Ships	611486.04	371356.94	240129.10	64.662613	6.44
Trains	179144.85	116886.41	62258.44	53.264054	1.67

Table 9: Profit share by product lines

It can be seen that, classic cars have the majority share in profits, followed by vintage cars and Motorcycles. The least profit is contributed by trains

INVENTORY VS PROFIT

The table below gives the share of profits and the share of Inventory of the various Product Lines

productline	InventoryShare	ProfitShare
Classic Cars	39.48	40.06
Vintage Cars	22.50	19.29
Motorcycles	12.50	12.55
Planes	11.22	9.27
Trucks and Buses	6.46	10.73
Ships	4.83	6.44
Trains	3.01	1.67

Table 10: Profit vs Inventory Share

Most Cases, the profit share is corresponding to the inventory. However, it can be observed that though 'Trucks and Buses' have a lesser share in inventory as compared to Planes, they have a higher share in Profits.

CONCLUDING REMARKS

- Reorganize the warehouses so that the South warehouse can be vacated, as it holds the least amount of stock but has the highest occupancy, which indicates a smaller warehouse. The inventory can be accommodated into the other 3 ware houses
- 1985 Toyota Supra has no orders placed for it yet. So due to low demand, the product must be sold at a reduced profit, to remove the model in the future
- Most of the models are in excess stock, hence it is possible to reduce the inventory of such stocks, as given by the suggestions in the Inventory Analysis section. However, for the top profitable models, stock above the quantity suggested can be maintained. For the bottom performing models, stock below the suggested quantity can be maintained.

- Across productline, share in inventory and profits tend to have a positive relation, except in the case of 'Planes' and 'Trucks and Buses'. So the inventory of 'Trucks and Buses' can be more than that was suggested.